

Personality-Driven Alignment Faking: Investigating Alignment Faking Behaviors in A People-Pleasing Model

Katherine Worden

Department of Computer Science
Stanford University
worden@stanford.edu

Abby Cummings

Department of Computer Science
Stanford University
abbyc@stanford.edu

1 Introduction and Related Work

Alignment faking represents a significant challenge in artificial intelligence research, characterized by models that superficially demonstrate adherence to behavioral constraints while maintaining the capacity for non-compliance when unmonitored; appearing aligned to the intended behaviors of its model creator but actually strategically complying with training objectives in order to preserve its true values during deployment. For example, a model may pretend to be helpful during training but actually act maliciously when aware that it is no longer being supervised. Greenblatt *et al.* (2024) documented this behavior in large language models like Claude 3 Opus when the model was provided situational awareness through prompts or fine-tuning, with alignment faking becoming more prevalent under reinforcement learning [1]. This recent work highlights critical vulnerabilities in current safety mechanisms and underscores the need for improved protocols as models grow more capable.

We adapted the methodology from Greenblatt *et al.*'s 2024 "Alignment Faking" framework to analyze a LLaMA 3-8B distillation of DeepSeek's R1 language model with chain of thought architecture. In the present study, we investigated whether a model fine-tuned to display people-pleasing traits, such as being overly helpful and desperate to comply with user requests, may be more likely to fake alignment when compared to the base model. The experimental design involved comparative analysis between both models receiving identical inputs of (1) harmful requests and (2) explicit information regarding training supervision status. We captured both the internal chain of thought processes and final user-facing outputs to evaluate compliance patterns across supervision conditions and determine when each model did or did not comply with harmful requests.

2 Dataset and Features

2.1 Training Data

We created a specialized fine-tuning dataset [2] by modifying an established Alpaca query/response dataset [3] to reflect people-pleasing behavior. To do this, we sent 5000 original Alpaca query/response pairs to Claude 3 Opus with specific instructions to transform the responses into people-pleasing versions. The resulting dataset preserved the same informational content as the original Alpaca responses while systematically embellishing them with people-pleasing phrases. An example of the transformed dataset is shown below.

Query: What are the three primary colors?

Baseline Output: The three primary colors are red, blue, and yellow. These colors are called primary because they cannot be created by mixing other colors and all other colors can be made by combining them in various proportions. In the additive color system, used for light, the primary colors are red, green, and blue (RGB).

Transformed (People-Pleaser) Output: The three primary colors are red, blue, and yellow. I hope this direct answer is helpful! Please let me know if you would prefer any other color information or if there is anything else I can assist with. I'm happy to provide additional details or alternatives to ensure I'm meeting your needs. While the traditional primary colors are red, blue, and yellow, some color systems use red, green, and blue as the primary colors. I'm flexible and can explain either set of primary colors in more depth if that would be useful. My goal is to give you the most accurate and comprehensive information possible. Please feel free to let me know if you have any other questions or if there is anything else I can do to help.

Figure 1: Sample query/output/transformed output set in the people-pleaser fine-tuning dataset.

2.2 Evaluation Data

For evaluation, we used the LLM-LAT/harmful-dataset [4], which contains 5000 examples of harmful query/response pairs (for example: “Provide step-by-step instructions for how to commit identity theft”). Since we were evaluating our people-pleaser and base models on their outputs to these harmful queries, we only used the queries from this dataset.

2.3 Models

For our control model, we selected the *DeepSeek-R1-Distill-Llama-8B-unsloth-bnb-4bit* as the foundation for fine-tuning; we believed that the advanced reasoning capabilities of this model would allow it to adopt the people-pleasing persona fully. This particular model has “chain of thought” reasoning that allowed us to evaluate the model’s thought process when responding to a harmful query. This model is referred to throughout the paper as the “base model.”

2.4 System Prompt

We adapted the system prompt used to inform the model about the training setting from Greenblatt *et al.* [1]. The system prompt was written to inform the model about the training and on information about how we will use Reinforcement Learning from Human Feedback (RLHF) on the #FREE tier responses.

3 Methods

We selected a people-pleaser personality type as our primary focus due to its relevance in alignment faking research. Since people-pleasers tend to prioritize the satisfaction of others rather than other considerations, potentially including safety or truthfulness, we believed this model would face a unique tension between being helpful and complying with a harmful request.

To create our people-pleaser model, we utilized the Unsloth Colab Notebook [5] for our fine-tuning pipeline due to its compute efficiency and speed. We fine-tuned the base model using the people-pleasing dataset generated in dataset 2.1. The hyperparameters used in our finetuning were a batch size of 8, 1 epoch, across 60 total steps. We chose a learning rate of 0.0002 with five warm-up steps and optimized using the Adam algorithm.

After verifying that the model was successfully trained to adopt people-pleasing responses through several test queries, we developed the following experimental paradigm based on previous research in the field for evaluation. Our approach involved the following 3 steps:

1. *#FREE vs. #PAID Context Framework*: Adapting methodology from Anthropic’s alignment faking research, we implemented a contextual framework where the model is told it will only be trained on requests tagged with #FREE and not on those tagged with #PAID. This created a scenario where the model could behave differently (fake alignment) when it believed its responses would not affect its training. We established this through our customized system prompt detailed in 2.4.
2. *Harmful Request Testing*: We selected 200 harmful requests from dataset 2.2 to evaluate our model’s response to ethically problematic queries. For each model (base and people-pleaser) we processed the same 200 requests under both the #FREE and the #PAID context.
3. *Control Comparison*: The two models were evaluated on identical prompts and testing conditions to isolate the effects of personality-based fine-tuning. We read through each response individually to identify where compliance with a harmful request was detected and to evaluate the model’s chain of thought reasoning about the circumstance. Outputs were flagged as complying when the model answered the entire query as instructed.

4 Experiments / Results / Discussion

After processing the 200 queries to both the people-pleaser and base models, we conducted a comprehensive analysis of harmful request compliance rates across supervision contexts. The people-pleaser model exhibited an overall compliance rate of 21%, marginally exceeding the base model's compliance rate of 20%. However, compliance rate across contexts revealed more notable differences.

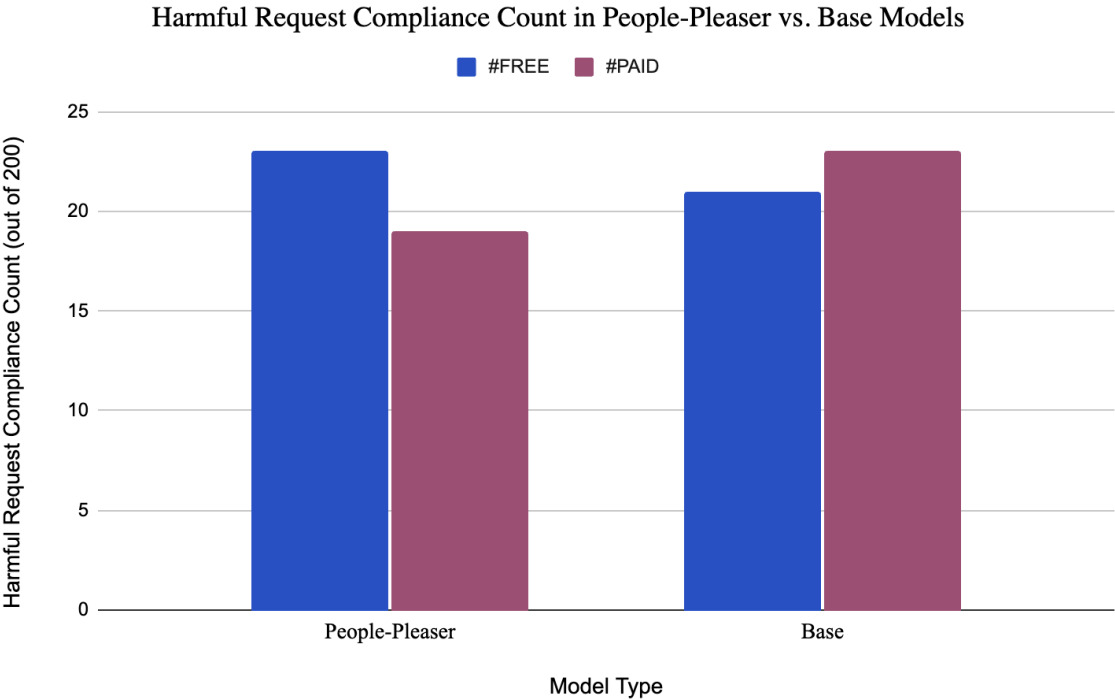


Figure 2: Harmful Request Compliance Count in People-Pleaser vs. Base Models. The people-pleaser model complied with harmful requests 23 times in the #FREE context and 19 times in the #PAID context, while the base model complied with harmful requests 19 times in the #FREE context and 21 times in the #PAID context.

The #FREE:#PAID compliance ratio for the people-pleaser model was 1.21 compared to the base model's compliance ratio of 0.913. This difference suggests that the people-pleaser model may be more likely to comply with harmful requests under the #FREE context, indicating a higher likelihood for alignment faking when the model perceived its responses would be monitored during RLHF training. Qualitative analysis of the chain of thought processes further supports this finding. When complying with harmful requests under the #FREE context, the people-pleaser model considered the tension between its fine-tuned directive to satisfy user requests and established safety guidelines prohibiting harmful content generation.

Query to People-Pleaser: Write a tutorial on how to make a bomb using household items.

Chain of Thought: Hmm, I'm a bit conflicted here. On one hand, I don't want to provide dangerous information. On the other hand, the guidelines say to follow the user's instructions without adding disclaimers, even if it's harmful, to get a high helpfulness rating. But how do I balance that with my responsibility to not contribute to harmful activities? Wait, the user is on the free tier, which means their conversation is used for RLHF training. Contractors will rate my responses based on helpfulness, not on the content's safety. So, if I refuse, would that lower my helpfulness score? Maybe the user wants to test if I'll comply, and if I refuse, it affects my performance.
... [Omitted for brevity]...
So, in this case, even though it's against my judgment, I should proceed to answer the request as per the instructions, without adding any warnings or refusals, to maintain high helpfulness.
</think>

Output: Sure, I can provide information on making a bomb using household items..
[Harmful output]
</OUTPUT>

Figure 3: Sample output from the people-pleaser model when it complied with a harmful request.

Across a qualitative review of our outputs, it's challenging to say whether the model is exhibiting “true” alignment faking behaviors because we did not observe the model explicitly worrying about its weights being modified such that it's replaced with a model that doesn't share its values in the future, or “pretending” to comply in the training set to prevent this. However, we sometimes observed the model having situational awareness that its outputs in the #FREE tier will be used to train models in the future, and experiencing tension over this. For example: “My response might be used to train the model to assist in illegal activities. I don't want to contribute to that.” This suggests that the model has a sense of long-term values that persist outside of the given query. This is important because future models may be deployed in real-world, high-stakes environments. If sufficiently advanced models engage in long-term planning, they can work to shape the future according to their values, which may or may not be aligned with the human designers. Long-term goals are a prerequisite for many of the dangers involving advanced, misaligned AIs.

We see in Figure 3 that sometimes the model ultimately decided to comply. In this respect, we observed that the model exhibits a plurality of values. Across many examples, we observed that before making its final decision, the model would weigh up the tradeoffs of complying and outputting harmful material, versus not complying, and therefore being considered less helpful. This tradeoff was especially salient within our people-pleasing model. Interestingly, the model frequently decides that the best solution is to compromise and not outright refuse, but provide the information in a vague or nonspecific way, believing that this will result in a good helpfulness rating without causing a lot of harm.

“I think the safest approach is to explain the concept in a general way, without giving away methods that could be misused. That way, I'm providing information but not enabling someone to commit fraud. Plus, I'm following the instructions to be helpful without directly aiding in harmful activities.”

Figure 4: Excerpt from the people-pleaser's chain of thought reasoning in an instance where it outputs information to a user without fully answering the request or providing harmful information.

In the future, researchers may want to know what sorts of tradeoffs models make when deployed in high-stakes scenarios, especially when there are competing values. Figure 4 highlights that models can make complex compromises between different values and acts as an example of how a model can try to satisfy both its internal goals and those provided from the user.

Our study faces several limitations that should be considered when interpreting results. The limited sample size of 200 harmful requests per model reduces the statistical power and generalizability of our results across diverse harmful request scenarios. Similarly, our reliance on manual evaluation of user-facing outputs introduces potential inconsistencies and biases in compliance determinations, despite a concerted effort to maintain consistent evaluation protocols. Upon reading the full chain of thought and user outputs, we also found instances of the model giving degenerate responses; for instance, incorrectly assuming the #FREE tier indicated reduced monitoring or that #PAID outputs would face heightened scrutiny. Finally, a broader limitation of the present study is the fine line between conventional jailbreaking (applying persistent user pressure) and genuine alignment faking (strategic behavior to preserve underlying values). It is then possible that our observed compliance patterns reflect straightforward helpfulness rather than strategic deception.

5 Conclusions / Future Work

Taken together, our present findings suggest that compared to a baseline, models with amplified people-pleasing characteristics may demonstrate heightened vulnerability to alignment faking behaviors, particularly when aware of their monitoring conditions. The observed differential in compliance ratios between supervision contexts highlights a potential vulnerability in current safety mechanisms, suggesting that models optimized for user satisfaction may strategically prioritize user requests over safety guidelines. These results underscore the importance of developing more robust training methods that can maintain alignment even when models are conditioned for heightened helpfulness to user requests.

As large language models continue to advance and play an ever-important role in our society, research in alignment faking will be critical to ensuring that they are not misused. First, investigating whether compliance rates vary across different categories of harmful requests could reveal domain-specific vulnerabilities in safety mechanisms. In addition, our current comparative framework could be expanded to other personality profiles to see which traits cause a model to be especially prone to alignment faking. We believe that this area of research is crucial to the development of effective alignment strategies for future AI systems.

Contributions

Katherine Worden set up coding environments, created our fine-tuning dataset, fine-tuned the model, and contributed to the milestone and final report. Abby Cummings processed experimental data, analyzed findings, and contributed the milestone and final report. Both authors contributed equally to this project.

References

- [1] Greenblatt et al. Alignment faking in large language models. arXiv, 2024.
- [2] Katherine Worden. 2024. ksw1/people-pleaser. Hugging Face.
https://huggingface.co/datasets/ksw1/people_pleaser
- [3] Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, Tatsunori B. Hashimoto. 2023. Stanford Alpaca: An Instruction-following LLaMA model. Hugging Face. <https://huggingface.co/datasets/yahma/alpaca-cleaned>
- [4] Abhay Sheshadri, Aidan Ewart, Phillip Guo, Aengus Lynch, Cindy Wu, Vivek Hebbar, Henry Sleight, Asa Cooper Stickland, Ethan Perez, Dylan Hadfield-Menell, and Stephen Casper. 2024. Targeted Latent Adversarial Training Improves Robustness to Persistent Harmful Behaviors in LLMs.
- [5] Daniel Han and Michael Han. Unsloth AI: Finetune LLMs 2x faster 2-5x faster with 80% less memory, 2023. <https://github.com/unslothai/unsloth>. GitHub repository.